Newton-Raphson Initialize

Algorithm

1. start
    a. Set n=1
    b. Set xr=x0
    c. Set er=1
2. Start the iteration loop:
    a. Condition: Continue if er> epes (error is greater than tolerance) and n≤ steps (iteration count is less than or equal to maximum steps).
3. Check the derivative:
    a. If df(xr)≠0 (i.e., the derivative is not zero), proceed to the next step.
4. Apply the Newton-Raphson update:
    i. Xr_new=xr$-f(xr)/df(xr)$
5. Update error:
    i. $er = | xr_new - xr/xr_new |$
6. Update iteration variables:
    a. Set xr=xr_new
    b. Increment iteration counter: n=n+1.
7. Check the stopping condition:
    a. If er≤ epes (error is smaller than tolerance), stop and return the current value of xr as the root.
    b. If n>steps (maximum steps reached), stop and return the current approximation with a warning about the maximum iterations.
8. Return the root:
    a. Return xr as the approximate root of the function.

Example

Find the root of $f(x)=x^2-4$ using the Newton-Raphson method.

Solution

1. Define $f(x)=x^2-4f$, so $f'(x)=2x$
2. Choose an initial guess, say $x0=3$.
3. Apply the iteration formula:

   Xr=x0

   Xr=3

   F(xr)=3*3-4=5 f'(xr)=2*3=6

   $$Xr_new=xr-f(xr)/df(xr)$$

   Xr_new=3-5/6

   =3-0.8333

   =2.1667

   Now update xr=xr_new

   Xr=2.1667

   Then, again find the root value using new xr